# ADVANCED

Stage 6

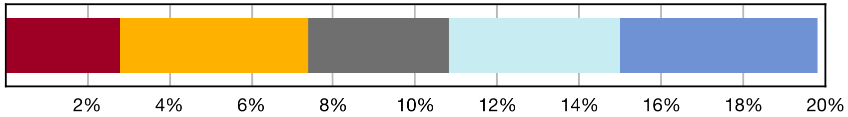
## Calculus (Adv), C3 Applications of Calculus (Adv)

### Curve Sketching (Y12)

**Teacher:** Kristy Chen

**Exam Equivalent Time:** 168 minutes *(based on allocation of 1.5 minutes per mark)*

**C3 Applications of Calculus**




The Derivative Function and its Graph
Curve Sketching
Tangents
Maxima and Minima
Rates of Change

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

## HISTORICAL CONTRIBUTION

- *C3 Applications of Calculus* is the 800 pound gorilla in the Advanced rainforest, contributing a massive 19.8% to new syllabus exams since introduced in 2020.
- This topic has been split into five sub-topics for analysis purposes: *1-The Derivative Function and its Graph (2.8%), 2-Curve Sketching (4.6%), 3-Tangents (3.4%), 4-Maxima and Minima (4.2%) and 5-Rates of Change (4.8%).*
- This analysis looks at *Curve Sketching.*

## HSC ANALYSIS - What to expect and common pitfalls

- *Curve Sketching (4.6%)* is consistently examined with significant mark allocations.
- The most popular curve is easily the cubic, asked 6 times in the last decade, including each year in the period 2017-2020.
- Important: The *2022 Adv 22* cubic asked for the "global maximum and minimum" within a given range for the first time. This required finding the $y$-values of the graph at the domain limits - a must review question!
- Sketches of polynomials of degree 4 are the second most popular, most recently tested in 2024 (see *2024 Adv 19* and *2012 Adv 14a*).
- *Curve Sketching* questions regularly require students to look at concavity and intervals where a curve is increasing or decreasing. An important revision area that proiduces low mean marks.

## Questions

### 1. Calculus, 2ADV C3 2011 HSC 7a

*Part i: RAP Data (vs state mean) - Bottom 21%: School result (77%) was 6% below state average (83%)*
*Part ii: RAP Data (vs state mean) - Bottom 9%: School result (69%) was 13% below state average (82%)*

Let $f(x) = x^3 - 3x + 2$ .

i. Find the coordinates of the stationary points of $y = f(x)$, and determine their nature. *(3 marks)*

ii. Hence, sketch the graph $y = f(x)$ showing all stationary points and the $y$-intercept. *(2 marks)*

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### 2. Calculus, 2ADV C3 2005 HSC 4b

A function $f(x)$ is defined by $f(x) = (x+3)(x^2-9)$ .

i. Find all solutions of $f(x) = 0$ *(2 marks)*

ii. Find the coordinates of the turning points of the graph of $y = f(x)$, and determine their nature. *(3 marks)*

iii. Hence sketch the graph of $y = f(x)$, showing the turning points and the points where the curve meets the $x$-axis. *(2 marks)*

iv. For what values of $x$ is the graph of $y = f(x)$ concave down? *(1 mark)*

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### 3. Calculus, 2ADV C3 2006 HSC 5a

A function $f(x)$ is defined by $f(x) = 2x^2(3-x)$ .

i. Find the coordinates of the turning points of $y = f(x)$ and determine their nature. *(3 marks)*

ii. Find the coordinates of the point of inflection. *(1 mark)*

iii. Hence sketch the graph of $y = f(x)$, showing the turning points, the point of inflection and the points where the curve meets the $x$-axis. *(3 marks)*

iv. What is the minimum value of $f(x)$ for $-1 \le x \le 4$? *(1 mark)*

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### 4. Calculus, 2ADV C3 2007 HSC 6b

Let $f(x) = x^4 - 4x^3$.

i. Find the coordinates of the points where the curve crosses the axes. *(2 marks)*

ii. Find the coordinates of the stationary points and determine their nature. *(4 marks)*

iii. Find the coordinates of the points of inflection. *(1 mark)*

iv. Sketch the graph of $y = f(x)$, indicating clearly the intercepts, stationary points and points of inflection. *(3 marks)*

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## 5. Calculus, 2ADV C3 2015 HSC 13c

Consider the curve $y = x^3 - x^2 - x + 3$.

i. Find the stationary points and determine their nature. *(4 marks)*

ii. Given that the point $P\left(\frac{1}{3}, \frac{70}{27}\right)$ lies on the curve, prove that there is a point of inflection at $P$. *(2 marks)*

iii. Sketch the curve, labelling the stationary points, point of inflection and $y$-intercept. *(2 marks)*

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## 6. Calculus, 2ADV C3 2016 HSC 13a

Consider the function $y = 4x^3 - x^4$.

i. Find the two stationary points and determine their nature. *(4 marks)*

ii. Sketch the graph of the function, clearly showing the stationary points and the $x$ and $y$ intercepts. *(2 marks)*

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## 7. Calculus, 2ADV C3 2017 HSC 13b

Consider the curve $y = 2x^3 + 3x^2 - 12x + 7$.

i. Find the stationary points of the curve and determine their nature. *(4 marks)*

ii. Sketch the curve, labelling the stationary points. *(2 marks)*

iii. Hence, or otherwise, find the values of $x$ for which $\dfrac{dy}{dx}$ is positive. *(1 mark)*

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## 8. Calculus, 2ADV C3 2018 HSC 13a

Consider the curve $y = 6x^2 - x^3$.

i. Find the stationary points and determine their nature. *(3 marks)*

ii. Given that the point (2,16) lies on the curve, show that it is a point of inflection. *(2 marks)*

iii. Sketch the curve, showing the stationary points, the point of inflection and the $x$ and $y$ intercepts. *(2 marks)*

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## 9. Calculus, 2ADV C3 2020 HSC 16

Sketch the graph of the curve $y = -x^3 + 3x^2 - 1$, labelling the stationary points and point of inflection. Do NOT determine the $x$-intercepts of the curve. *(4 marks)*

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## 10. Calculus, 2ADV C3 2024 HSC 19

Sketch the curve $y = x^4 - 2x^3 + 2$ by first finding all stationary points, checking their nature, and finding the points of inflection. *(5 marks)*

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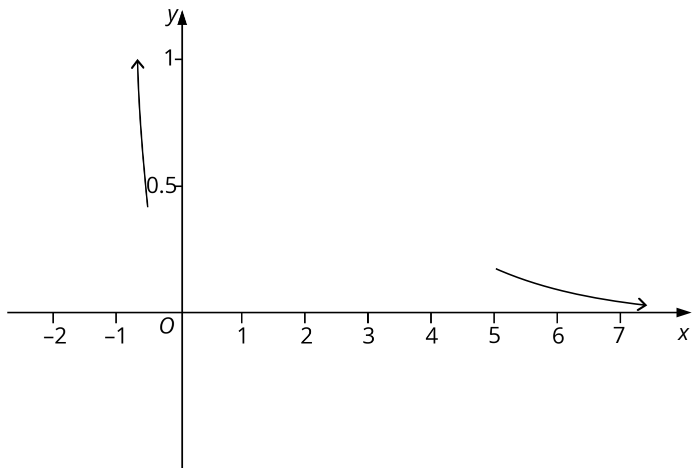
## 11. Calculus, 2ADV C3 2025 HSC 16

Consider the function $f(x) = \dfrac{x^2}{e^x}$.

a. Find the stationary points of the function and determine their nature. *(4 marks)*

b. A partially completed graph of $f(x) = \dfrac{x^2}{e^x}$ is shown.

Use your answer from part (a) to complete the graph. *(1 mark)*

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## 12. Calculus, 2ADV C3 2022 HSC 22

Find the global maximum and minimum values of $y = x^3 - 6x^2 + 8$, where $-1 \le x \le 7$. *(4 marks)*

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## 13. Calculus, 2ADV C3 2019 HSC 14b

The derivative of a function $y = f(x)$ is given by $f'(x) = 3x^2 + 2x - 1$.

a. Find the $x$-values of the two stationary points of $y = f(x)$, and determine the nature of the stationary points. *(2 marks)*

b. The curve passes through the point $(0, 4)$.
Find an expression for $f(x)$. *(2 marks)*

c. Hence sketch the curve, clearly indicating the stationary points. *(2 marks)*

d. For what values of $x$ is the curve concave down? *(1 mark)*

## 14. Calculus, 2ADV C3 2004 HSC 4b

Consider the function $f(x) = x^3 - 3x^2$.

i. Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. *(3 marks)*

ii. Sketch the curve showing where it meets the axes. *(2 marks)*

iii. Find the values of $x$ for which the curve $y = f(x)$ is concave up. *(2 marks)*

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## 15. Calculus, 2ADV C3 2010 HSC 6a

Let $f(x) = (x+2)(x^2+4)$.

i. Show that the graph $y = f(x)$ has no stationary points. *(2 marks)*

ii. Find the values of $x$ for which the graph $y = f(x)$ is concave down, and the values for which it is concave up. *(2 marks)*

iii. Sketch the graph $y = f(x)$, indicating the values of the $x$ and $y$ intercepts. *(2 marks)*

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## 16. Calculus, 2ADV C3 2008 HSC 8a

Let $f(x) = x^4 - 8x^2$.

i. Find the coordinates of the points where the graph of $y = f(x)$ crosses the axes. *(2 marks)*

ii. Show that $f(x)$ is an even function. *(1 mark)*

iii. Find the coordinates of the stationary points of $f(x)$ and determine their nature. *(4 marks)*

iv. Sketch the graph of $y = f(x)$. *(1 mark)*

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## 17. Calculus, 2ADV C3 2012 HSC 14a

*Part iv: RAP Data (vs state mean) - Bottom 10%: School result (0%) was 12% below state average (12%)*
*Part iv: RAP Data (raw) - School result was 0%*

A function is given by $f(x) = 3x^4 + 4x^3 - 12x^2$.

i. Find the coordinates of the stationary points of $f(x)$ and determine their nature. *(3 marks)*

ii. Hence, sketch the graph $y = f(x)$ showing the stationary points. *(2 marks)*

iii. For what values of $x$ is the function increasing? *(1 mark)*

iv. For what values of $k$ will $f(x) = 3x^4 + 4x^3 - 12x^2 + k = 0$ have no solution? *(1 mark)*



# Worked Solutions



## 1. Calculus, 2ADV C3 2011 HSC 7a

**i.**

$$f(x) = x^3 - 3x + 2$$
$$f'(x) = 3x^2 - 3$$
$$f''(x) = 6x$$

Stationary points when $f\prime(x) = 0$

$$3x^2 - 3 = 0$$
$$3(x^2-1) = 0$$
$$\therefore x^2 = 1$$
$$x = \pm 1$$

When $x = 1$

$$f(1) = 1 - 3 + 2 = 0$$
$$f''(1) = 6 > 0$$

$\therefore$ MIN S.P. at $(1, 0)$

When $x = -1$

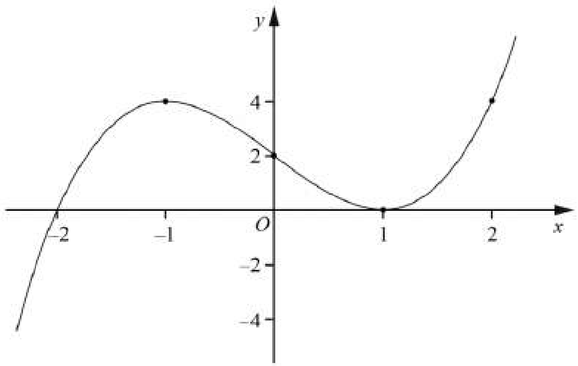
$$f(-1) = -1 + 3 + 2 = 4$$
$$f''(-1) = -6 < 0$$

$\therefore$ MAX S.P. at $(-1, 4)$

**ii.** $y = x^3 - 3x + 2$

$y$-intercept $= 2$

**MARKER'S COMMENT:** Graphs should be *large* (around ½ page), axes drawn *with a ruler*, with intercepts and turning points clearly shown. The scale can be different on each axe for clarity, as shown in the Worked Solution.



## 2. Calculus, 2ADV C3 2005 HSC 4b

**i.**
$$\begin{aligned} f(x) &= (x+3)\left(x^2-9\right) \\ &= (x+3)(x+3)(x-3) \end{aligned}$$

$\therefore f(x) = 0$ when $x = -3$ or $3$

**ii.**
$$\begin{aligned} f(x) &= (x+3)\left(x^2-9\right) \\ &= x^3 - 9x + 3x^2 - 27 \\ &= x^3 + 3x^2 - 9x - 27 \end{aligned}$$

$$f\prime(x) = 3x^2 + 6x - 9$$

$$f\prime\prime(x) = 6x + 6$$

S.P.'s when $f\prime(x) = 0$

$$\begin{aligned} 3x^2 + 6x - 9 &= 0 \\ 3\left(x^2 + 2x - 3\right) &= 0 \\ 3(x-1)(x+3) &= 0 \end{aligned}$$

At $x = 1$

$$f(1) = (4)(-8) = -32$$

$$f\prime\prime(1) = 6 + 6 = 12 > 0$$

$\therefore$ MIN at $(1, -32)$

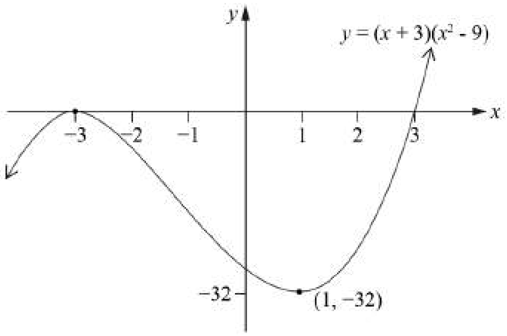
At $x = -3$

$$f(-3) = 0$$

$$f\prime\prime(-3) = (6 \times -3) + 6 = -12 < 0$$

$\therefore$ MAX at $(-3, 0)$

**iii.**




**iv.** $f(x)$ is concave down when

$$\begin{aligned} f\prime\prime(x) &< 0 \\ 6x + 6 &< 0 \\ 6x &< -6 \\ x &< -1 \end{aligned}$$

3. Calculus, 2ADV C3 2006 HSC 5a

**i.** $f(x) = 2x^2(3-x)$

$= 6x^2 - 2x^3$

$f'(x) = 12x - 6x^2$

$f''(x) = 12 - 12x$

S.P.'s when $f'(x) = 0$

$12x - 6x^2 = 0$

$6x(2-x) = 0$

$x = 0 \text{ or } 2$

When $x = 0$

$f(0) = 0$

$f''(0) = 12 - 0 = 12 > 0$

$\therefore$ MIN at $(0, 0)$

When $x = 2$

$f(2) = 2 \times 2^2(3-2) = 8$

$f''(2) = 12 - (12 \times 2) = -12 < 0$

$\therefore$ MAX at $(2, 8)$

**ii.** P.I. when $f''(x) = 0$

$12 - 12x = 0$

$12x = 12$

$x = 1$

$f''(0.5) = 6 > 0$

$f''(1.5) = -6 < 0$

Since concavity changes $\Rightarrow$ P.I. exists

$f(1) = 2 \times 1^2(3-1)$

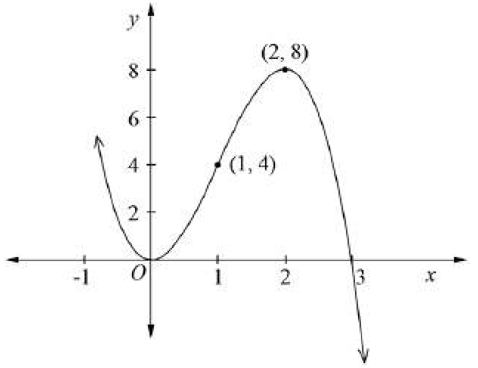
$= 4$

$\therefore$ P.I. at $(1, 4)$

**iii.** $f(x)$ meets $x$-axis when $f(x) = 0$

$2x^2 \times (3-x) = 0$

$x = 0 \text{ or } 3$




**(iv)** The graph clearly shows that in the given range $-1 \le x \le 4$, the minimum will occur when $x = 4$

$\therefore$ Minimum $= 2 \times 4^2(3-4)$

$= -32$

## 4. Calculus, 2ADV C3 2007 HSC 6b

**i.** $f(x) = x^4 - 4x^3$

Cuts $x$-axis when $f(x) = 0$

$$x^4 - 4x = 0$$
$$x^3(x-4) = 0$$
$$x = 0 \text{ or } 4$$

$\therefore$ Cuts the $x$-axis at $(0,0)$ , $(4,0)$

Cuts the $y$-axis when $x = 0$

$\therefore$ Cuts the $y$-axis at $(0,0)$

**ii.** $f(x) = x^4 - 4x^3$

$f'(x) = 4x^3 - 12x^2$

$f''(x) = 12x^2 - 24x$

S.P.'s when $f'(x) = 0$

$$4x^3 - 12x^2 = 0$$
$$4x^2(x-3) = 0$$
$$x = 0 \text{ or } 3$$

When $x = 0$

$f(0) = 0$

$f''(0) = 0$

| $x$ | -0.5 | 0 | 0.5 |
|---|---|---|---|
| $f''(x)$ | $>0$ | 0 | $<0$ |

Since concavity changes, a P.I. occurs at $(0,0)$

When $x = 3$

$$f(3) = 3^4 - 4 \times 3^3$$
$$= -27$$

$$f''(3) = 12 \times 3^2 - 24 \times 3$$
$$= 36 > 0$$

$\therefore$ Minimum S.P. at $(3, -27)$

**iii.** P.I. when $f''(x) = 0$

$$12x^2 - 24x = 0$$
$$12x(x-2) = 0$$
$$x = 0 \text{ or } 2$$

P.I. at $(0,0)$ (from(ii))

When $x = 2$

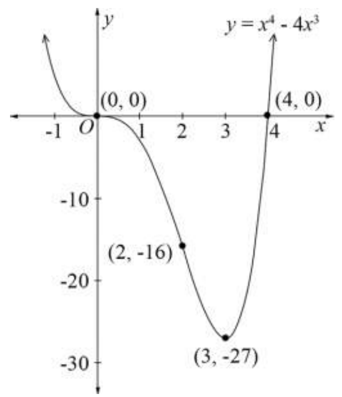
| $x$ | 1.5 | 2 | 2.5 |
|---|---|---|---|
| $f''(x)$ | $<0$ | 0 | $>0$ |

Since concavity changes, a P.I. occurs when $x = 2$

$$f(2) = 2^4 - 4 \times 2^3$$
$$= 16$$

$\therefore$ P.I.'s at $(0,0)$ and $(2,16)$

**iv.**



5. Calculus, 2ADV C3 2015 HSC 13c

**i.** $y = x^3 - x^2 - x + 3$

$$\frac{dy}{dx} = 3x^2 - 2x - 1$$

$$\frac{d^2y}{dx^2} = 6x - 2$$

S.P.'s when $\dfrac{dy}{dx} = 0$

$$3x^2 - 2x - 1 = 0$$
$$(3x+1)(x-1) = 0$$
$$x = -\frac{1}{3} \text{ or } 1$$

When $x = -\dfrac{1}{3}$

$$f\left(-\frac{1}{3}\right) = \left(-\frac{1}{3}\right)^3 - \left(-\frac{1}{3}\right)^2 - \left(-\frac{1}{3}\right) + 3$$
$$= -\frac{1}{27} - \frac{1}{9} + \frac{1}{3} + 3$$
$$= \frac{86}{27}$$

$$f''\left(-\frac{1}{3}\right) = \left(6 \times -\frac{1}{3}\right) - 2 = -4 < 0$$

$\therefore$ MAX at $\left(-\dfrac{1}{3}, \dfrac{86}{27}\right)$

When $x = 1$

$$f(1) = 1^3 - 1^2 - 1 + 3 = 2$$
$$f''(1) = (6 \times 1) - 2 = 4 > 0$$

$\therefore$ MIN at $(1, 2)$

**ii.** $\dfrac{d^2y}{dx^2} = 0$ when

$$6x - 2 = 0$$
$$x = \frac{1}{3}$$

Checking change of concavity

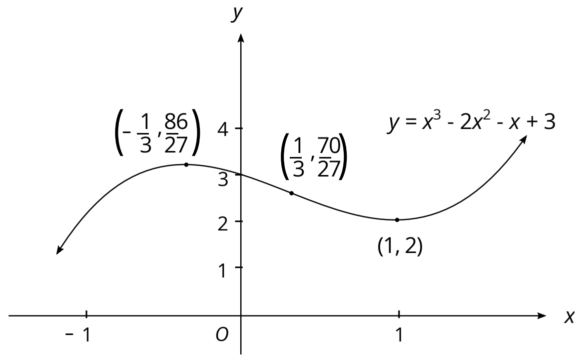
| $x$ | $\frac{1}{6}$ | $\frac{1}{3}$ | $\frac{1}{2}$ |
|---|---|---|---|
| $f(x)$ | $-1 < 0$ | 0 | $1 > 0$ |

Concavity changes either side of $x = \dfrac{1}{3}$

$\therefore$ $\left(\dfrac{1}{3}, \dfrac{70}{27}\right)$ is a P.I.

**iii.** When $x = 0$

$y = 3$



## 6. Calculus, 2ADV C3 2016 HSC 13a

**i.**

$$y = 4x^3 - x^4$$
$$y\prime = 12x^2 - 4x^3$$
$$y\prime\prime = 24x - 12x^2$$

S.P.'s when $y\prime = 0$,

$12x^2 - 4x^3 = 0$

$4x^2(3 - x) = 0$

$\therefore x = 0$ or $3$

When $x = 0$, $y\prime\prime(0) = 0$

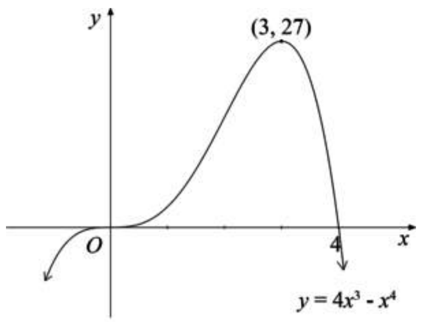
$\therefore$ P.I. at $(0, 0)$

When $x = 3$,

$y\prime\prime(3) = 24(3) - 12(9) = -36 < 0$

$\therefore$ MAX at $(3, 27)$

**ii.**

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## 7. Calculus, 2ADV C3 2017 HSC 13b

**i.**

$$y = 2x^3 + 3x^2 - 12x + 7$$
$$\frac{dy}{dx} = 6x^2 + 6x - 12$$
$$\frac{d^2y}{dx^2} = 12x + 6$$

S.P. when $\dfrac{dy}{dx} = 0$

$6x^2 + 6x - 12 = 0$

$x^2 + x - 2 = 0$

$(x + 2)(x - 1) = 0$

$x = -2$ or $1$

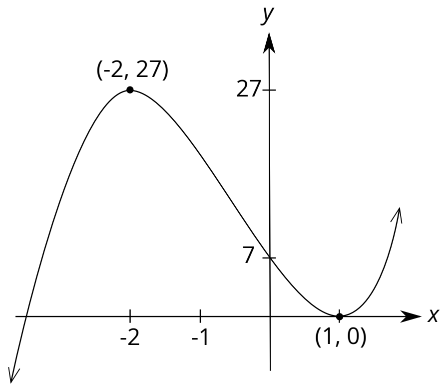
When $x = -2$, $\dfrac{d^2y}{dx^2} < 0$

$\therefore$ MAX at $(-2, 27)$

When $x = 1$, $\dfrac{d^2y}{dx^2} > 0$

$\therefore$ MIN at $(1, 0)$

**ii.**




**iii.** Solution 1

From graph, gradient is positive for

$x < -2$ and $x > 1$

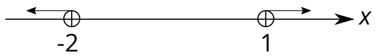
$\therefore \dfrac{dy}{dx} > 0$ for $x < -2$ and $x > 1$

Solution 2

$$\frac{dy}{dx} > 0$$

$$6x^2 + 6x - 12 > 0$$

$$(x+2)(x-1) > 0$$




$\therefore x < -2$ and $x > 1$

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## 8. Calculus, 2ADV C3 2018 HSC 13a

**i.** $y = 6x^2 - x^3$

$$\frac{dy}{dx} = 12x - 3x^2$$

$$\frac{d^2y}{dx^2} = 12 - 6x$$

S.P.s occur when $\dfrac{dy}{dx} = 0$

$$12x - 3x^2 = 0$$

$$3x(4-x) = 0$$

$$x = 0 \text{ or } 4$$

When $x = 0$, $\dfrac{d^2y}{dx^2} > 0$

$\therefore$ MIN at $(0, 0)$

When $x = 4$, $\dfrac{d^2y}{dx^2} < 0$

$\therefore$ MAX at $(4, 32)$

**ii.** P.I. occur when $\dfrac{d^2y}{dx^2} = 0$,

$$12 - 6x = 0$$

$$x = 2$$

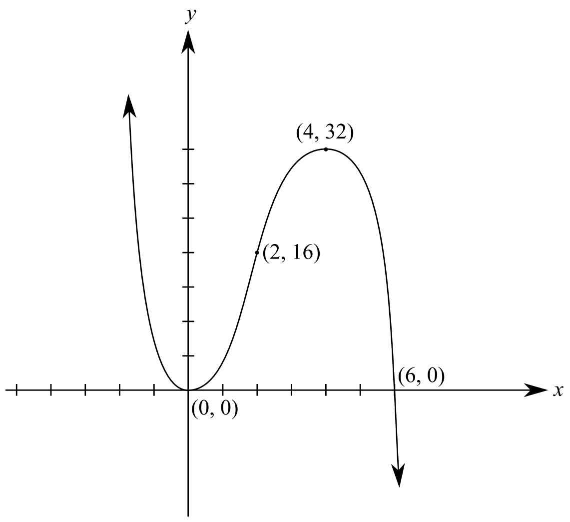
When $x = 2$, $y = 16$

| $x$ | $2^-$ | $2$ | $2^+$ |
|---|---|---|---|
| $\dfrac{d^2y}{dx^2}$ | $>0$ | $0$ | $<0$ |

Since the concavity changes

$\Rightarrow$ P.I. occurs at $(2, 16)$

**iii.**




9. Calculus, 2ADV C3 2020 HSC 16

$$y = -x^3 + 3x^2 - 1$$

$$\frac{dy}{dx} = -3x^2 + 6x$$

$$\frac{d^2y}{dx^2} = -6x + 6$$

SP's when $\dfrac{dy}{dx} = 0$

$$-3x^2 + 6x = 0$$

$$-3x(x-2) = 0$$

$$x = 0 \text{ or } 2$$

When $x = 0$,

$$y = -1$$

$$\frac{d^2y}{dx^2} = 6 > 0$$

$\therefore$ MIN at $(0, -1)$

When $x = 2$,

$$y = -8 + 12 - 1 = 3$$

$$\frac{d^2y}{dx^2} = -6 \times 2 + 6 = -6 < 0$$

$\therefore$ MAX at $(2, 3)$

$\dfrac{d^2y}{dx^2} = 0$ when

$$-6x + 6 = 0$$

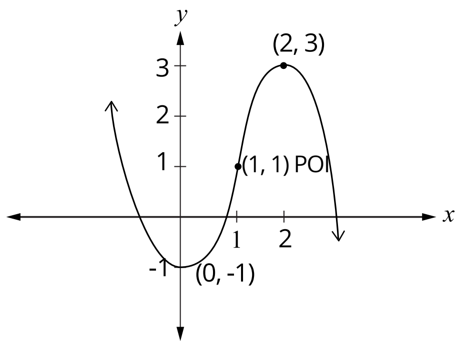
$$x = 1$$

Checking change of concavity

| $x$ | $\frac{1}{2}$ | 1 | $\frac{3}{2}$ |
|---|---|---|---|
| $\frac{d^2y}{dx^2}$ | 3 | 0 | −3 |

Concavity changes either side of $x = 1$

$\therefore$ POI at $(1, 1)$




10. Calculus, 2ADV C3 2024 HSC 19

$$y = x^4 - 2x^3 + 2$$
$$y' = 4x^3 - 6x^2 = 2x^2(2x-3)$$
$$y'' = 12x^2 - 12x = 12x(x-1)$$

SP's when $y' = 0$:

$2x^2(2x-3) = 0 \;\Rightarrow\; x = 0 \text{ or } \dfrac{3}{2}$

At $x = 0,\ y'' = 0$

| $x$ | $-1$ | $0$ | $1$ |
|---|---|---|---|
| $y'$ | $-10$ | $0$ | $-5$ |

$\Rightarrow$ Horizontal POI at $(0, 2)$

At $x = \dfrac{3}{2}$:

$$y'' = 12 \times \frac{3}{2}\left(\frac{3}{2} - 1\right) = 9 > 0, \quad y = \left(\frac{3}{2}\right)^4 - 2\left(\frac{3}{2}\right)^3 + 2 = \frac{5}{16}$$

$\Rightarrow$ MIN at $\left(\dfrac{3}{2}, \dfrac{5}{16}\right)$

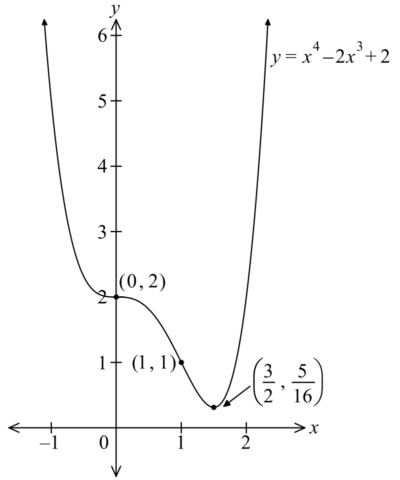
POI when $y'' = 0$:

$12x(x-1) = 0 \;\Rightarrow\; x = 1 \text{ or } x = 0$ (see above)

Test concavity change at $x = 1$:

| $x$ | $\frac{1}{2}$ | $1$ | $\frac{3}{2}$ |
|---|---|---|---|
| $y''$ | $-3$ | $0$ | $9$ |

$\Rightarrow$ POI at $(1, 1)$




11. Calculus, 2ADV C3 2025 HSC 16

**a.** $f(x) = \dfrac{x^2}{e^x}$

$$
\begin{aligned}
f'(x) &= 2x \cdot e^x - e^x \cdot x^2 \\
&= \frac{xe^x(2-x)}{e^{2x}} \\
&= \frac{x(2-x)}{e^x}
\end{aligned}
$$

Find $x$ when $f'(x) = 0$:

$x(2-x) = 0$

$x = 0$ or $2$

When $x = 0 \Rightarrow f(0) = 0$

When $x = 2 \Rightarrow f(2) = \dfrac{4}{e^2}$

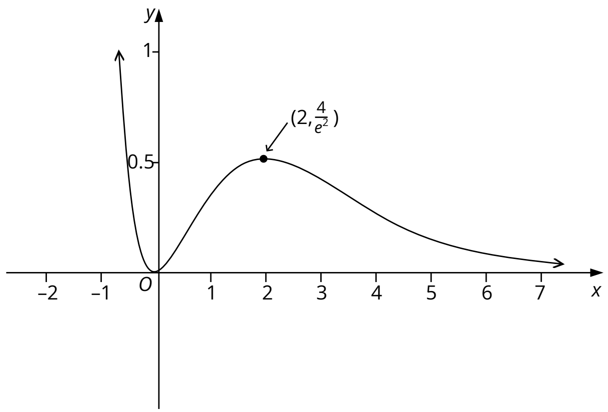
Checking nature of SP's:

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|
| $f'(x)$ | $-3e$ | $0$ | $\dfrac{1}{e}$ | $0$ | $-\dfrac{3}{e^3}$ |

$\therefore$ MIN at $(0, 0)$

MAX at $\left(2, \dfrac{4}{e^2}\right)$

**b.**



## 12. Calculus, 2ADV C3 2022 HSC 22

$$y = x^3 - 6x^2 + 8$$

$$\frac{dy}{dx} = 3x^2 - 12x$$

$$\frac{d^2y}{dx^2} = 6x - 12$$

SP's when $\dfrac{dy}{dx} = 0$:

$3x^2 - 12x = 0$

$3x(x-4) = 0$

$x = 0$ or $4$

When $x = 0, \; y = 8, \; \dfrac{d^2y}{dx^2} < 0$

$\Rightarrow$ Local Max at $(0, 8)$

When $x = 4, \; y = 4^3 - 6(4^2) + 8 = -24, \; \dfrac{d^2y}{dx^2} > 0$

$\Rightarrow$ Local Min at $(4, -24)$

Check ends of domain:
When $x = -1, \; y = -1 - 6 + 8 = 1$
When $x = 7, \; y = 7^3 - 6(7^2) + 8 = 57$
$\therefore$ Global max $= 57$
$\therefore$ Global min $= -24$

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## 13. Calculus, 2ADV C3 2019 HSC 14b

**a.** $f'(x) = 3x^2 + 2x - 1$

$f''(x) = 6x + 2$

S.P.'s when $f'(x) = 0$

$3x^2 + 2x - 1 = 0$

$(3x-1)(x+1) = 0$

$x = \dfrac{1}{3}$ or $-1$

When $x = \dfrac{1}{3}$,

$f''(x) = 4 > 0 \Rightarrow$ MIN

When $x = -1$,

$f''(x) = -4 < 0 \Rightarrow$ MAX

**b.** $f(x) = \displaystyle\int f'(x)\,dx$

$= \displaystyle\int 3x^2 + 2x - 1\,dx$

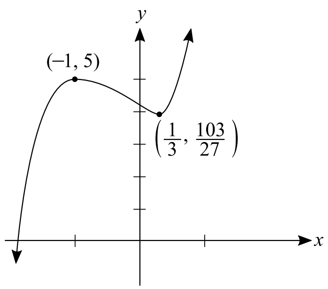
$= x^3 + x^2 - x + c$

$(0, 4)$ lies on $f(x) \Rightarrow c = 4$

$\therefore f(x) = x^3 + x^2 - x + 4$

**c.** When $x = -1, \; y = 5$

When $x = \dfrac{1}{3}, \; y = \dfrac{103}{27}$




**d.** Concave down when $f''(x) < 0$

$6x + 2 < 0$

$$6x < -2$$
$$x < -\frac{1}{3}$$

*Mean mark 36%.*

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## 14. Calculus, 2ADV C3 2004 HSC 4b

**(i)** $f(x) = x^3 - 3x^2$

$f'(x) = 3x^2 - 6x$

$f''(x) = 6x - 6$

S.P.'s when $f'(x) = 0$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$x = 0 \text{ or } 2$

When $x = 0$

$f(0) = 0$

$f''(0) = 0 - 6 = -6 < 0$

$\therefore$ MAX at $(0, 0)$

When $x = 2$

$f(2) = 2^3 - (3 \times 4) = -4$

$f''(2) = (6 \times 2) - 6 = 6 > 0$

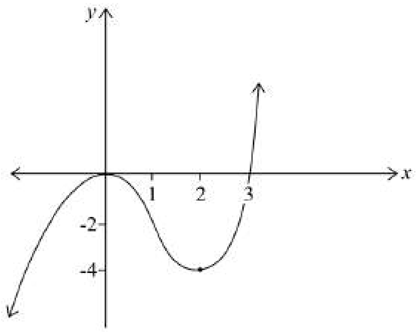
$\therefore$ MIN at $(2, -4)$

**(ii)** $f(x) = x^3 - 3x^2$ meets the $x$-axis when $f(x) = 0$

$x^3 - 3x^2 = 0$

$x^2(x - 3) = 0$

$x = 0 \text{ or } 3$




**(iii)** $f(x)$ is concave up when

$f''(x) > 0$

$6x - 6 > 0$

$6x > 6$

$x > 1$

$\therefore\ f(x)$ is concave up when $x > 1$

---

## 15. Calculus, 2ADV C3 2010 HSC 6a

**i.** Need to show no S.P.'s

$$f(x) = (x+2)\left(x^2+4\right)$$
$$= x^3 + 2x^2 + 4x + 8$$
$$f\prime(x) = 3x^2 + 4x + 4$$

S.P.s occur when $f\prime(x) = 0$,

$3x^2 + 4x + 4 = 0$

$$\Delta = b^2 - 4ac$$
$$= 4^2 - (4 \times 3 \times 4)$$
$$= 16 - 48$$
$$= -32 < 0$$

Since $\Delta < 0$, No Solution

$\therefore$ No S.P.'s for $f(x)$

**ii.** $f(x)$ is concave down when $f\prime\prime(x) < 0$

$f\prime\prime(x) = 6x + 4$

$$\Rightarrow 6x + 4 < 0$$
$$6x < -4$$
$$x < -\frac{2}{3}$$

$\therefore\ f(x)$ is concave down when $x < -\dfrac{2}{3}$

$f(x)$ is concave up when $f\prime\prime(x) > 0$

$f\prime\prime(x) = 6x + 4$

$$\Rightarrow 6x + 4 > 0$$
$$6x > -4$$
$$x > -\frac{2}{3}$$

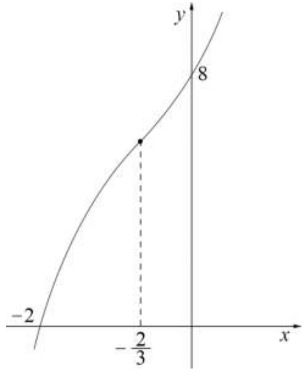
$\therefore\ f(x)$ is concave up when $x > -\dfrac{2}{3}$

**MARKER'S COMMENT:** The significance of the sign of the second derivative was not well understood by most students.

**iii.** $y$-intercept $= 2 \times 4 = 8$

$x$-intercept $= -2$

*Mean mark 33%.*
**MARKER'S COMMENT:** Students are reminded to bring a ruler to the exam and use it to draw the axes for graphing and to help with an appropriate scale.




16. Calculus, 2ADV C3 2008 HSC 8a

**i.** $f(x) = x^4 - 8x^2$

$y$-intercept when $x = 0$

$\therefore$ Cuts $y$ axis at $(0, 0)$

$x$ -intercept when $f(x) = 0$

$$x^4 - 8x^2 = 0$$
$$x^2(x^2 - 8) = 0$$
$$x^2 - 8 = 0 \quad \text{or} \quad x = 0$$
$$x^2 = 8$$
$$x = \pm\sqrt{8} = \pm 2\sqrt{2}$$

$\therefore$ Cuts $x$-axis at $\left(-2\sqrt{2}, 0\right)$, and $\left(2\sqrt{2}, 0\right)$

(Note that it only touches $x$-axis at (0,0))

**ii.** 
$$f(x) = x^4 - 8x^2$$
$$f(-x) = (-x)^4 - 8(-x)^2$$
$$= x^4 - 8x^2$$
$$= f(x)$$

$\therefore$ $f(x)$ is an even function.

**iii.** 
$$f(x) = x^4 - 8x^2$$
$$f'(x) = 4x^3 - 16x$$
$$f''(x) = 12x^2 - 16$$

S.P. when $f'(x) = 0$

$$4x^3 - 16x = 0$$
$$4x(x^2 - 4) = 0$$
$$x^2 - 4 = 0 \quad x = 0$$
$$x^2 = 4$$
$$x = \pm 2$$

At $x = 2$ $\quad f(x) = (2)^4 - 8(2)^2 = -16$

$$f''(x) = 12(2^2) - 16 > 0$$

$\therefore$ MIN at (2, –16)

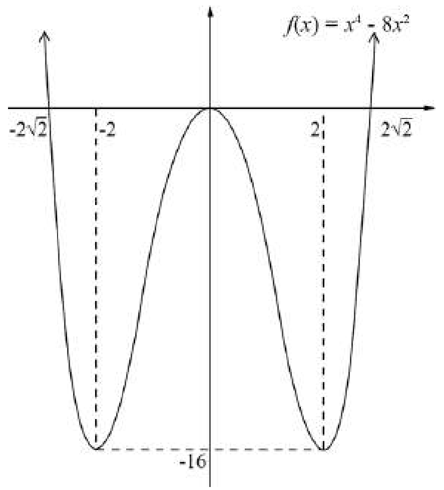
$\therefore$ MIN at (–2, –16), $\quad$ ($f(x)$ is even)

At $x = 0$ $\quad f(x) = (0)^4 - 8(0)^2 = 0$

$f''(x) = 12(0^2) - 16 < 0$

$\therefore$ MAX at (0,0)

**iv.**




17. Calculus, 2ADV C3 2012 HSC 14a

**i.** $f(x) = 3x^4 + 4x^3 - 12x^2$

$f'(x) = 12x^3 + 12x^2 - 24x$

$f''(x) = 36x^2 + 24x - 24$

Stationary points when $f'(x) = 0$

$12x^3 + 12x^2 - 24x = 0$

$12x(x^2 + x - 2) = 0$

$12x(x+2)(x-1) = 0$

$\therefore$ Stationary points at $x = 0,\ 1$ or $-2$

When $x = 0$, $\quad f(0) = 0$

$f''(0) = -24 < 0$

$\therefore$ MAX at (0,0)

When $x = 1$

$f(1) = 3 + 4 - 12 = -5$

$f''(1) = 36 + 24 - 24 = 36 > 0$

$\therefore$ MIN at $(1, -5)$

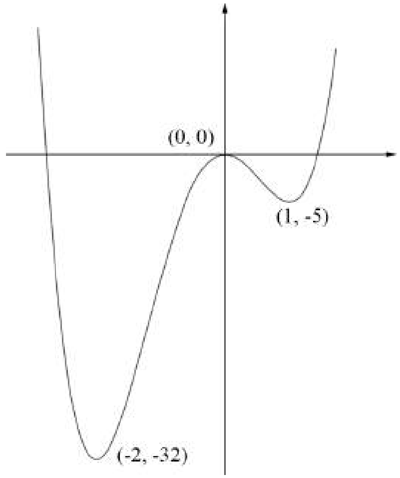
When $x = -2$

$$f(-2) = 3(-2)^4 + 4(-2)^3 - 12(-2)^2$$
$$= 48 - 32 - 48$$
$$= -32$$

$$f''(-2) = 36(-2)^2 + 24(-2) - 24$$
$$= 144 - 48 - 24 = 72 > 0$$

$\therefore$ MIN at $(-2, -32)$

**ii.**




*Mean mark 42%*
**MARKER'S COMMENT:** Be careful to use the correct inequality signs, and not carelessly include ≥ or ≤ by mistake.

**iii.** $f(x)$ is increasing for

$-2 < x < 0$ and $x > 1$

**iv.** Find $k$ such that

*Mean mark 12%.*

$3x^4 + 4x^3 - 12x^2 + k = 0$ has no solution

$k$ is the vertical shift of $y = 3x^4 + 4x^3 - 12x^2$

$\Rightarrow$ No solution if it does not cross the $x$-axis.

$\therefore$ No solution when $k > 32$